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Systematic Review

Doula Support During Pregnancy, Childbirth, and Postpartum

A Systematic Review and Meta-Analysis

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OBJECTIVE: To systematically review the evidence on doula support in pregnancy, birth, and postpartum.

DATA SOURCES: PubMed, Cochrane Library (which includes clinical trials), CINAHL, PsycInfo.

METHODS OF STUDY SELECTION: Eligible studies were English-language randomized controlled trials (RCTs) and nonrandomized studies of interventions (NRSIs) based in the United States that evaluated doula support compared with standard care. Included studies reported maternal, infant, and partner outcomes from database inception through November 18, 2025. We used GRADE (Grading of Recommendations Assessment, Development and Evaluation) to assess the certainty of evidence.

TABULATION, INTEGRATION, AND RESULTS: Twelve RCTs and 18 NRSIs reported on doula support compared with standard maternity care. One study had low risk of bias (ROB) (one RCT), 10 had moderate ROB (three RCTs, seven NRSIs), and 19 had high ROB (eight RCTs, 11 NRSIs). Doula support is likely to be associated with lower rates of preterm birth and cesarean delivery, improved birth experiences, and increased likelihood of postpartum visit attendance (moderate certainty of evidence). Doula support may be associated with improved prenatal mood, spontaneous vaginal birth, vaginal birth after cesarean, initiation of breastfeeding, childbirth education attendance, respectful care, satisfaction with care, self-efficacy, parenting confidence, postpartum adjustment to parenthood, and reduced epidural anesthesia use, length of labor, and infant mortality (low certainty of evidence). It may not be associated with reduced rates of hypertensive disorders of pregnancy, maternal readmissions, and neonatal intensive care admissions (low certainty of evidence). Among studies composed entirely of people at risk of disparities (specifically, those arising from race, carceral settings, low income, and having public insurance), four of five reported at least some benefit associated with doula sup-

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port. There was insufficient information available on the effects of varying intensity or types of doula support. Heterogeneity was noted across many outcomes.

CONCLUSION: Research suggests that doula support is associated with some improved maternal outcomes, although many studies were heterogeneous and rated as having high ROB. Research on type (including independent doulas), intensity, and timing of doula support could help to clarify settings in which doula care improves maternal and child outcomes and reduces health disparities.

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The rate of maternal mortality in the United States far exceeds the average rate in other highly developed regions (21 deaths vs 6–11 deaths/100,000 live births in Europe and Canada in 2020).[1] Maternal mortality in the United States is further characterized by disparities based on race and ethnicity, education, income, and rural compared with urban location.[2–4] Disparities in maternal health outcomes may be amplified when women experience multiple disadvantages.[5]

Much of the birth-related cultural knowledge of communities now at risk for increased rates of maternal mortality was lost when obstetric care transitioned primarily to hospital settings.[6] Doula support has been proposed to reduce maternal and infant health disparities. Specifically,[7–9] doulas who share patients' culture may offer a potential way to reduce health disparities[10,11] and mitigate negative outcomes influenced by interpersonal and institutional biases[12,13] through enhanced transparency and communication between clinicians and patients. Doulas are nonclinical trained professionals who advocate for patients and provide "emotional, physical, and informational support during pregnancy, delivery, and after childbirth."[7,14–17]

This systematic review investigated the following key questions:

1. What is the effect of doula support on maternal, infant, and partner outcomes compared with standard maternity care without doula support?
2. Does doula support that includes the prenatal period, postpartum period, or both lead to better maternal, infant, and partner outcomes than doula support only offered during childbirth?
3. Does the effect of doula support during the prenatal period, postpartum period, or both vary by doula support characteristics, specifically type of doula training, skills, and experience; doula type; doula employer; intensity of care (timing of support, number of interactions, length of interactions, type of interactions [number and type of resources and referrals, follow-up on referrals]); mode of interaction (virtual vs in person); and race, language, or cultural concordance?

SOURCES

Our protocol for this systematic review was registered on April 29, 2024, in the Open Science Framework and updated on November 13, 2025.[18] The protocol reflects revisions, comments received from Technical Expert Panel members, and public comments after posting on the PCORI (Patient-Centered Outcomes Research Institute) website from March 1 to March 22, 2024.

The review was informed by guidance from the National Academy of Medicine, Agency for Healthcare Research and Quality, PCORI Methodology Standards for Systematic Reviews, Cochrane, and the Equity Rubric.[19–23] Detailed information on these methodology standards is available in Appendix 1 (available online at https://links.lww.com/AOG/E807).

Our inclusion and exclusion criteria (Table 1 includes condensed criteria; Appendix 2, available online at https://links.lww.com/AOG/E807, reports detailed criteria) were used to guide searches in PubMed, Cochrane Library (which includes trials from ClinicalTrials.gov), CINAHL, and PsycInfo (Appendix 3, available online at https://links.lww.com/AOG/E807) for English-language articles reporting studies through November 18, 2025. We included gray literature publications (ie, clinical trial records, conference abstracts, dissertations) in database searches and conducted targeted searches of doula organization and doula services websites (eg, National Black Doulas Association). We limited the review to the United States given its relatively unique landscape of birth in hospitals and known disparities. We manually searched the reference lists of relevant recent systematic reviews to identify additional citations. We included randomized controlled trials (RCTs) and prospective or retrospective nonrandomized studies of interventions (NRSIs) that compared standard maternity care plus doula support with standard maternity care without doula support or an active comparator for pregnant, birthing, or postpartum individuals. The selected studies reported on a variety of maternal, infant, or partner outcomes determined a priori to be potentially affected by doula support (see Table 1).

We included studies for each outcome only if the doula support was initiated before the outcome. For

Table 1. Inclusion and Exclusion Criteria

Category	Inclusion Criteria[*]	Exclusion Criteria
Population	• Pregnant, birthing, or postpartum women	• Individuals who were not pregnant or recently pregnant
Interventions	• KQ 1–3. Doula care[†] overall, timing of doula care, and doula care characteristics including training, intensity, mode of interaction, or race, language, or cultural concordance	• Support provided by a relative, partner, friend, untrained or unskilled supporter, or health care professional • Doula support exclusively focused on abortion, sex, and death
Comparators	• KQ 1. Standard maternity care without doula support • KQ 2. Doula care during childbirth alone • KQ 3. Doula care characteristics	• Studies without a comparator
Outcomes (relevant for all KQs)[‡]		
Maternal outcomes		
Birth intervention and outcomes	• Adverse pregnancy outcomes • Mode of delivery • Labor outcomes • Perinatal mood, anxiety, substance use, and mental health disorders • Health care access and use	• Outcomes not listed as eligible
Postpartum outcomes	• Length of skin-to-skin contact • Length of time that neonate was with mother before separation • Lactation and breastfeeding or pumping outcomes • Postpartum mood, anxiety, substance use, and mental health disorders • Health care access and use	• Outcomes not listed as eligible
Postpartum patient-reported outcomes	• Birth experience • Reports of respectful care or discrimination, mistreatment, or racism by health care team • Satisfaction with care (doula or health care team) • Shared decision making • Self-esteem and self-efficacy • Parenting confidence • Postpartum adjustment	• Outcomes not listed as eligible
Infant outcomes	• Apgar score • Growth parameters • Delayed cord clamping • Neonatal complications • NICU or hospital utilization • Discharge of neonate with family member(s) • Neonatal and infant mortality	• Outcomes not listed as eligible
Partner outcomes	• Birth experience • Reports of respectful care or discrimination, mistreatment, or racism by health care team • Satisfaction with care (doula or health care team) • Self-esteem and self-efficacy • Parenting confidence • Postpartum adjustment	• Outcomes not listed as eligible
Patient harms	• Any reported safety or harms outcomes	• NA
Timing	• Prenatal, during labor and childbirth, up to 12 mo postpartum	• Before pregnancy, more than 1 y postpartum for maternal postpartum outcomes
Setting		
Geographic	• United States	• All other countries
Care delivery	• In-person or virtual doula support in any setting including email, text, phone	• NA

(*continued*)

Table 1. Inclusion and Exclusion Criteria (*continued*)

Category	Inclusion Criteria*	Exclusion Criteria
Study design	• RCTs, nonrandomized controlled trials, and observational comparative studies	• Any other study design
Language	• English	• Non-English

KQ, key question; NICU, neonatal intensive care unit; NA, not applicable; RCT, randomized controlled trial.
* See detailed inclusion and exclusion criteria in Appendix 2 (available online at https://links.lww.com/AOG/E807).
† Doulas included private doulas, community-based doulas, Indigenous doulas, and hospital doulas. Doula skills or training varied by study. The type of training or skills needed for inclusion was not prespecified.
‡ Only outcomes that were relevant for all comparison groups in a study were evaluated (eg, birth outcomes for a study comparing birth doulas with postpartum doulas were not eligible).

example, when evaluating the risk of hypertensive disorders of pregnancy, we included only those studies in which doula support began during antenatal care and excluded those in which doula support started in the hospital. We required that the studies describe support as offered by doulas or birth workers and relied on the existing definition of doula. Specifically, *doulas* are defined as trained or skilled individuals who offer educational, physical, or emotional support during pregnancy, delivery, or postpartum.[14–17] We did not prespecify the type of training or skills needed for inclusion but did not include untrained birth attendants.

STUDY SELECTION

We used DistillerSR[24] to screen articles and extract data; we tracked results in an EndNote bibliographic database. All abstracts were reviewed by at least two reviewers; any abstracts marked for inclusion were sent on to full-text review. Full-text articles were screened independently by two reviewers, with disagreements resolved by discussion. Excluded articles are reported in Appendix 4 (available online at https://links.lww.com/AOG/E807). Two reviewers also assessed risk of bias (ROB) and certainty of evidence; any discrepancies were also resolved by discussion.

To assess ROB, we used Cochrane's revised Risk of Bias 2 tool for RCTs[25] and the Risk Of Bias In Non-randomized Studies of Interventions (ROBINS-I)[26] for NRSIs. On the basis of cumulative answers to the tools' signaling questions, we assigned an overall ROB rating of low, moderate, or high to each study. Details of ROB assessments are presented in Appendices 5 and 6 (available online at https://links.lww.com/AOG/E807).

We used GRADE (Grading of Recommendations, Assessment, Development and Evaluation) to rate the certainty of evidence.[27] Certainty grades (high, moderate, low, very low) reflect our confidence in whether effect estimates across the body of evidence for an outcome are close to true effects. We assessed certainty of evidence separately for RCTs and NRSIs (detailed certainty of evidence in Appendix 7, available online at https://links.lww.com/AOG/E807).[28] We presented the analyses as follows: 1) outcomes with moderate certainty of evidence (from either RCTs or NRSIs or both), 2) outcomes with low certainty of evidence (from either RCTs or NRSIs or both), 3) outcomes with very low certainty of evidence, and 4) outcomes with no evidence. When certainty of evidence was not consistent between RCT and NRSI evidence, we focused on the body of evidence with greatest certainty, regardless of study design.[28] Outcomes with moderate certainty suggest, with moderate confidence, that the intervention is effective.[29] Outcomes graded as very low certainty (ie, very low confidence that the intervention is effective) or with no evidence point to opportunities for future research.

For the key questions, we extracted all included studies into tables (Appendix 8, available online at https://links.lww.com/AOG/E807). When at least three studies with similar interventions were available, we conducted random effects meta-analyses using the inverse-variance weighted method of DerSimonian and Laird in Comprehensive Meta-Analysis 3.3 to generate pooled estimates of effect.[30,31] Whenever possible, we generated risk differences for categorical data. For NRSIs, we pooled adjusted effect estimates (adjusted relative risks [RRs] or odds ratios [ORs]) whenever available and were therefore generally unable to generate risk differences. We based significance testing on the exclusion of the null value by the 95% CI around the pooled estimate. All testing was two-sided. We conducted sensitivity analyses with a random effects model estimated with restricted maximum likelihood in R (Appendix 9, available online at https://links.lww.com/AOG/E807). We performed additional analyses to assess the effect of ROB on the pooled estimates.

Appendix 7 includes detailed grading methods, and Appendix 9 includes full analytic methods (available online at https://links.lww.com/AOG/E807). We described key applicability considerations for all outcomes graded as at least low certainty of evidence in results tables (Appendix 10, available online at https://links.lww.com/AOG/E807). We summarized available information on populations at risk of health disparities for each key question (Appendix 11, available online at https://links.lww.com/AOG/E807).[32,33] Patients at risk for health disparities include, but are not limited to, those in racial and ethnic minority groups; rural-dwelling populations and those with low income; those identifying as LGBTQ+ (lesbian, gay, bisexual, transgender, queer+); those with low health or limited digital literacy; those with physical, intellectual, or developmental disabilities; those experiencing housing uncertainty or living in carceral settings; those who are immigrants or refugees; and those with limited English proficiency.[34]

RESULTS

We included 30 studies representing 12 RCTs (13 publications)[11,35–46] and 18 nonrandomized studies (from 23 publications, including one dissertation[47] that reported two studies) (Fig. 1).[12,47–68] Comparison groups specified included standard or usual care (n=14),[11,36–43,45–48,53,62,63] no doula (n=13),[44,49–52,54–59,64–68] or the active comparisons of case management[12,35] or Lamaze[60,61] (n=3). One study had low ROB (one RCT),[39] 10 had moderate ROB (three RCTs, seven NRSIs),[11,35,37,49,50,52,55–58,64,65] and 19 had high ROB (eight RCTs, 11 NRSIs).[12,36,38,40–48,51,53,54,59–63,68]

Included studies were heterogeneous in terms of populations and degree of doula support (Table 2, with additional details in Appendix 8, https://links.lww.com/AOG/E807). In 9 of 30 studies (30%), non-Hispanic White patient participants constituted a majority. African descent/non-Hispanic Black or

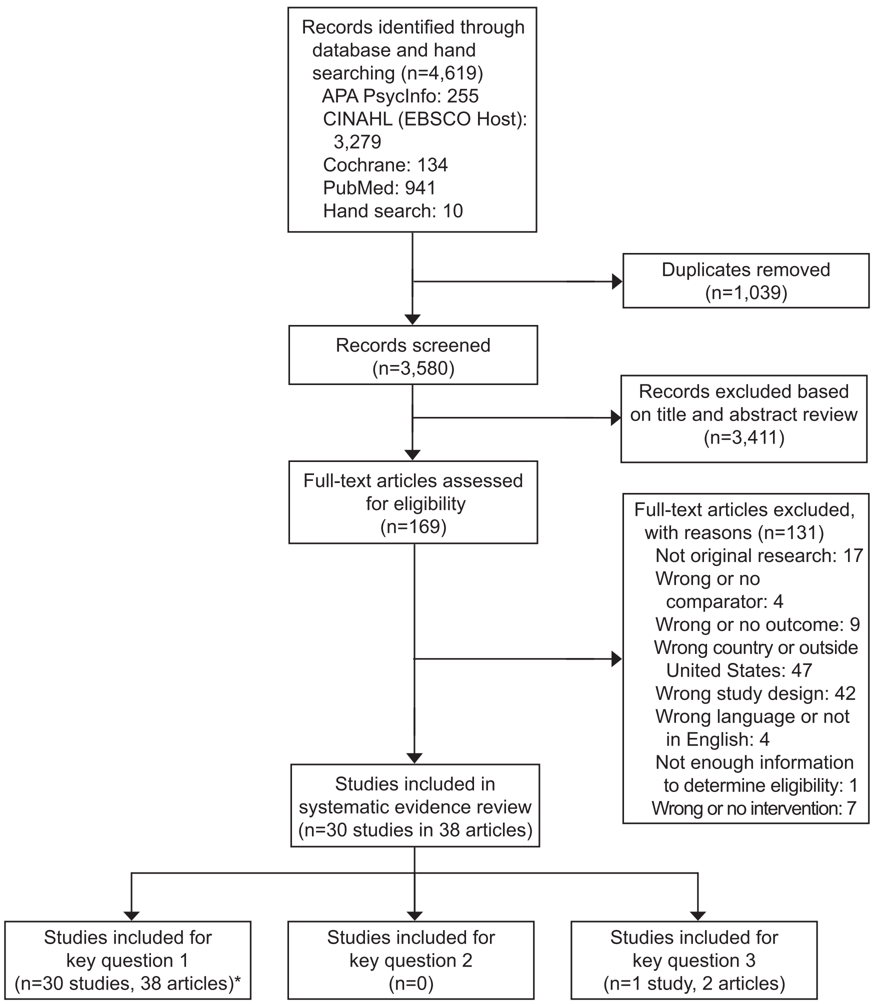

Fig. 1. PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) diagram. *Intervention participants in two included studies overlapped[56,63] (Everyday Miracles in Minneapolis, MN; January 1, 2010–April 30, 2012). The comparison included data from the 2009–2010 Minnesota Pregnancy Risk Assessment Monitoring System (PRAMS) for one study[63] and the 2009 Nationwide Inpatient Sample (NIS), Healthcare Cost and Utilization Project (HCUP) Agency for Healthcare Research and Quality for the other study.[56] There is some potential overlap between the comparison groups, but the extent is unclear. Due to this uncertainty, the articles were counted as two separate studies. One publication[51] provided evidence for two included studies: Maternal Care Study (MCS) II[51] and MCS I.[51,66,67]

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Table 2. Key Study Characteristics

Characteristic	Studies [n (%)][*]
Population characteristics[†,‡]	
Race and ethnicity	
Majority non-Hispanic White	9 (30.0)
Majority non-White[§]	19 (63.3)
NR	2 (6.7)
Age	
Adolescents and adults	18 (60.0)
Adults only	12 (40.0)
1st live birth	
1st birth only	9 (30.0)
Mixed	13 (43.3)
NR	8 (26.7)
Public insurance	
Majority receiving public insurance	11 (36.7)
Majority not receiving public insurance	4 (13.3)
NR	15 (50.0)
Immigrant status	
Majority immigrant status	1 (3.3)
Majority not immigrants	3 (10.0)
NR	26 (86.7)
Intervention characteristics	
Doula intervention type[‖]	
Hospital-based doula program	20 (66.7)
Virtual doula support	1 (3.3)
Hospital-based postpartum doula services	1 (3.3)
Home visiting and hospital based	1 (3.3)
Labor support only	1 (3.3)
Medicaid plan doula support	1 (3.3)
Lay doulas trained by DONA-trained social workers	1 (3.3)
Doula teams integrated in obstetric care	1 (3.3)
Nonprofit community doula organization	1 (3.3)
Unspecified	2 (6.7)
Timing	
Birth only	12 (40.0)
Postpartum only	1 (3.3)
Prenatal and birth only	2 (6.7)
Birth and postpartum only	1 (3.3)
Prenatal, birth, and postpartum	12 (40.0)
NR	1 (3.3)
Varied by patient	1 (3.3)
Mode of doula support delivery	
In person	19 (63.3)
Virtual	1 (3.3)
Combination	1 (3.3)
NR	9 (30.0)
Doula employer or supervisor	
Hospital or birth or medical center	6 (20.0)
Nonprofit or program	9 (30.0)
Research study	4 (13.3)
Insurance resource	1 (3.3)
NR	9 (30.0)
NA (lay volunteer)	1 (3.3)
Doula payment	
Unpaid volunteer	3 (10.0)
Unspecified	23 (76.7)
Specified	4 (13.3)

(*continued*)

Table 2. Key Study Characteristics (*continued*)

Characteristic	Studies [n (%)][*]
Doula characteristics	
Doula training	
DONA training or certification	8 (26.7)
Other doula training or certification	10 (33.3)
Lay doula	1 (3.3)
Family member or friend with training	1 (3.3)
NR	10 (33.3)
Doula race	
100% African descent/non-Hispanic Black	2 (6.7)
More than 50% non-White[‡]	3 (10.0)
More than 50% non-Hispanic White	1 (3.3)
NR	24 (80.0)
Cultural concordance	
Yes	7 (23.3)
NR	23 (76.7)
Language concordance	
Yes	5 (16.7)
NR	25 (83.3)
Described as community doula	
Yes	7 (23.3)
Unclear	4 (13.3)
NR	19 (63.3)
Other study characteristics	
Design	
RCT	12 (40.0)
NRSI	18 (60.0)
Comparator	
Usual care	14 (46.7)
No doula or control	13 (43.3)
Active[¶]	3 (10.0)
Recruitment setting	
Hospitals, clinic	17 (56.7)
Community programs	4 (13.3)
Claims, survey data, EMR	7 (23.3)
Prison	1 (3.3)
Various settings	1 (3.3)
Urbanicity	
Urban	5 (16.7)
NR	25 (83.3)
Risk of bias	
Low	1 (3.4)
Moderate	10 (33.3)
High	19 (63.3)
Funding	
Industry	1 (3.3)
No industry	20 (66.7)
Unfunded	3 (10.0)
NR	6 (20.0)

NR, not reported; DONA, Doulas of North America; NA, not applicable; RCT, randomized controlled trial; NRSI, nonrandomized studies of interventions; EMR, electronic medical record.
[*] Rounded to ensure that the total summed to 100%.
[†] No studies reported on the study characteristics of LGBTQ+ (lesbian, gay, bisexual, transgender, queer+), rural, literacy, or disabilities.
[‡] One study[11] reported on food, housing, or energy insecurity.
[§] Studies described participants as those of African descent, non-Hispanic Black, Hispanic/Latina, multiracial, Native American, Asian or Pacific Islander.
[‖] Doula intervention categories were mutually exclusive.
[¶] Three studies included the active comparators of case management[12,35] or Lamaze.[60,61]

Hispanic/Latina or other non-White (ie, multiracial, Native American, Asian or Pacific Islander) participants constituted the majority in 19 studies, and two studies did not report race or ethnicity. Thirty-seven percent of studies reported more than 50% of participants having public insurance; insurance coverage was not reported by 50% of studies.

Studies examined a variety of models for doula support, including hospital-based birth support, virtual support during labor and delivery, hospital-based postpartum support, home visiting plus birth support, birth support only, Medicaid-reimbursed support, support by lay doulas trained by social workers with DONA (Doulas of North America) certification, support by doulas integrated in obstetric teams, or unspecified support. More than half of the interventions reported that doulas were affiliated with a hospital or health system, and none explicitly reported using private, independently operating doulas. Almost a quarter of studies (23%) reported the use of "community doulas," though definitions of this designation varied widely.

Timing of doula support varied: 12 studies reported that doulas provided support throughout the entire perinatal period (ie, before birth, during birth, and postpartum), two only before and during birth, 12 only during birth, one only during birth and postpartum, and one only postpartum. In one study, support timing varied by patient; timing of support was not reported in one remaining study. The majority of studies used doulas who were trained, received certification from DONA, or both (eight studies) or an unspecified formal doula training or certification (10 studies). Five studies reported on doula language and cultural concordance with their clients,[11,12,37,52,53,57,59] and two studies reported on cultural concordance only.[11,37]

We sought evidence on 79 outcomes that our team and Technical Expert Panel members felt were plausibly influenced by doula support (Table 1) and found evidence on 35 outcomes that we could grade for certainty of evidence, which ranged from moderate to very low. We present the findings for these outcomes for moderate and low certainty of evidence (Table 3; forest plots for meta-analyses and sensitivity analyses detailed in Appendix 9, https://links.lww.com/AOG/E807). Sensitivity analyses were consistent with primary analyses.

Evidence from two moderate-ROB RCTs[11,35] provided inconsistent and imprecise results, leading to very low certainty of evidence of benefit from doula support provided before delivery. However, NRSI evidence from eight studies (five moderate ROB[52,55,56,58,64,67] and three high ROB[48,53,68]) offered moderate certainty of benefit (pooled OR for preterm birth: 0.73; 95% CI, 0.62–0.86, N=450,572; I^2: 53%; Appendix 9 Fig. 1, https://links.lww.com/AOG/E807). A sensitivity analysis removing high-ROB NRSIs also demonstrated an association with reduced risk of preterm birth (pooled OR 0.68; 95% CI, 0.56–0.83[52,55,56,58,64,67]; Appendix 9 Fig. 2, https://links.lww.com/AOG/E807).

Pooled estimates from eight RCTs (one low ROB,[39] two moderate ROB,[11,35] and five high ROB[38,40–44]) showed that women receiving doula support were less likely to have cesarean deliveries (16.0%, 229/1,433) when compared with those not receiving doula support (19.3%, 277/1,438) (pooled RD: −0.04; 95% CI, −0.074 to −0.005).[11,35,39–44] The findings from 15 NRSIs (five moderate ROB[50,55–58,64] and 10 high ROB[12,47–49,53,54,59,60,62,65]) also suggested reduced odds of cesarean delivery (pooled OR 0.70; 95% CI, 0.56–0.87).[12,47–50,52,55–57,60,62,64,65] Pooled estimates of cesarean delivery from RCTs for first births only were similar to those for all births. When high-ROB studies were removed from the meta-analysis, the difference in the risk of having a cesarean was similar to overall estimates, but the results were imprecise, with CIs spanning the null (risk difference: −0.04; 95% CI, −0.13 to 0.04[11,35,39]; Appendix 9 Figs. 3–7, https://links.lww.com/AOG/E807). The heterogeneity of the pooled NRSI analyses may be related to differences in study populations. Concerns arising from ROB limited certainty of evidence to moderate for RCTs; additionally, limitations arising from inconsistency in NRSIs limited certainty of evidence to low for NRSIs.

Birth experience measures varied but centered on the experience of birth rather than on satisfaction with doula care. Evidence from RCTs suggested moderate certainty of benefit,[38,41] whereas NRSI evidence suggested very low certainty of no benefit. Across all studies, measures of birth experience varied greatly (eg, rating of birth experience on a Likert scale,[38] feeling birth experience was good[41] satisfaction with birth [yes or no],[47,62,63] gratification with labor and delivery on a 104-item questionnaire[60,61]), precluding meta-analyses. Two high-ROB RCTs (N=809) reported that participants who received doula support were more likely to rate their birth experience as very good (RR 2.28; 95% CI, 1.80–2.88)[38] or good (RR 1.23; 95% CI, 1.08–1.40).[41] Three high-ROB NRSIs (N=235) did not find statistically significant differences in satisfaction with birth as assessed with Likert scale rating; estimates of effect showed wide CIs that spanned the null.[47,60–63]

Table 3. Certainty of Evidence for Graded Outcomes

COE and Outcome	Design	No. of Studies	No. of Participants	ROB
Moderate COE of benefit				
Adverse pregnancy outcomes				
Preterm birth	RCT	2	679	2 moderate[11,35]
	NRSI	8	450,572	5 moderate[52,55,56,58,64,67] 3 high[48,53,68]
Mode of delivery				
Cesarean	RCT	8	2,871	1 low[39] 2 moderate[11,35] 5 high[40–44]
	NRSI	15	397,843	5 moderate[50,55–58,64] 10 high[12,47–49,53,54,59,60,62,65]
Health care access and use				
Office visit within 6–12 wk postpartum	NRSI	2	19,254	2 moderate[64,67]
Patient-reported outcomes				
Birth experience	RCT	2	809	2 high[38,41]
	NRSI	3	235	3 high[47,60–63]
Low certainty of benefit or no benefit				
Adverse pregnancy outcomes				
HDP	RCT	1	367	1 moderate[11]
	NRSI	2	19.383	1 moderate[64] 1 high[65]
Mode of delivery				
SVD	RCT[‡]	4	1,232	1 moderate[11] 3 high[40,41,44]
	NRSI[§]	1	105	1 high[47,62,63]
VBAC	NRSI	2	4,589	2 moderate[64,67]
Use of obstetric intervention				
Epidural anesthesia	RCT	9	3,519	1 low[39] 2 moderate[11,35] 6 high[38,40,41,43,44,46]
	NRSI	3	11,756	1 moderate[57] 2 high[12,47]
Labor outcomes				
Overall length of labor; length of 2nd stage of labor	RCT	4	1,258	4 high[40,42,43,45]
Perinatal outcomes				
Prenatal mood	NRSI	1	8,989	1 moderate[66]
Postpartum outcomes				
Initiation of breastfeeding or pumping	RCT	4	1,187	3 moderate[11,35,37] 1 high[41]
	NRSI	4	64,627[‖]	1 moderate[57] 3 high[12,59,62,63,§]
Health care access and use				
Childbirth education attendance	RCT	1	312	1 moderate[35]
Maternal readmission within 30 d postpartum	RCT	1	312	1 moderate[35]
Maternal readmission within 6–12 wk postpartum	NRSI	3	20,827	2 moderate[64,67] 1 high[65]
Patient-reported outcomes				
Reports of respectful care (including feeling heard, autonomy, informed refusal)	NRSI	1	1,977	1 high[51]
Satisfaction with care (doula or health care team, received excellent care or high levels of support)	RCT	1	221	1 high[43]
	NRSI	1	8,989	1 moderate[50]
Self-efficacy (including confidence in navigating health care system)	NRSI	3	9,049	1 moderate[50,66] 2 high[47,60,61]
Parenting confidence	RCT	2	878	2 high[38,41]
Postpartum adjustment	RCT	1	564	1 high[38]

(*continued*)

Table 3. Certainty of Evidence for Graded Outcomes (*continued*)

COE and Outcome	Design	No. of Studies	No. of Participants	ROB
Infant outcomes				
NICU admission	RCT	2	679	2 moderate[11,35]
	NRSI	3	20,034	1 moderate[64] 2 high[45,48]
Infant mortality	RCT	1	312	1 moderate[35]
	NRSI	1	75,842	1 high[68]

COE and Outcome	Summary of Findings (Doula vs Standard Maternity Care)	Grade	Direction of Effect	Applicability Considerations
Moderate COE of benefit				
Adverse pregnancy outcomes				
Preterm birth	10/149 (6.7%) vs 12/146 (8.2%), OR 0.57, 95% CI, 0.22–1.46[35] 17/187 (9.1%) vs 13/180 (7.2%), OR 1.28, 95% CI, 0.61–2.73[11]	Very low	No benefit	Studies included varied populations, including those with low SES (having Medicaid or other public insurance, from high-poverty communities), incarcerated women, and refugees; varied doula services but also included a medical–legal partnership; 3 studies noted cultural concordance[11,52,53]
	All NRSIs: 588/8,372 (7.02%) vs 34,708/445,558 (7.27), pooled OR 0.73, 95% CI, 0.62–0.85, k=8, I^2: 53% Without high-ROB studies: 287/4,825 (5.94%) vs 26,631/364,010 (7.32%), pooled OR 0.68, 95% CI, 0.56–0.83, k=5, I^2: 51%[52,55,56,58,64,67]	Moderate	Benefit	
Mode of delivery				
Cesarean	229/1,433 (16.0%) vs 277/1,438 (19.3%), pooled RD −0.04; 95% CI, −0.074 to −0.005, I^2: 40.2% (Appendix 9 Figs. 3–6 for main and sensitivity analyses)*	Moderate	Benefit	Most studies were conducted in women with low SES; 5 studies reported cultural concordance[11,12,52,53,57]; doula support intensity varied: labor-only support,[41] labor support focused on pain reduction,[44] labor plus postpartum support,[43] prenatal, birth, and postpartum support plus medical–legal partnership with advocacy[11]
	Pooled OR 0.71, 95% CI, 0.58–0.87, I^2: 87%† (Appendix 9 Figs. 7–10 for main and sensitivity analyses)*	Low	Benefit	

(*continued*)

Table 3. Certainty of Evidence for Graded Outcomes (*continued*)

COE and Outcome	Summary of Findings (Doula vs Standard Maternity Care)	Grade	Direction of Effect	Applicability Considerations
Health care access and use				
Office visit within 6–12 wk postpartum	465/722 (64.6%) vs 322/722 (44.6%), PSM RR 1.46, 95% CI, 1.31–1.61[67] 404/486 (83.1%) vs 13,559/17,324 (78.3%), PS-adjusted RR 1.09, 95% CI, 1.05–1.13)[64]	Moderate	Benefit	1 study was hospital-based from a single institution with hospital-employed doulas[64]; the other was based on Medicaid claims from 9 states[67]
Patient-reported outcomes				
Birth experience	Overall rating of birth experience was very good: 134/229 (59%) vs 68/265 (26%), calculated RR 2.28, 95% CI, 1.80–2.88[38] Felt the birth experience was good: 123/149 (82.5%) vs 111/165 (67.4%), *P*=.005, RR 1.23, 95% CI, 1.08–1.40[41]	Moderate	Benefit	1 study was of women with public insurance[47]; interventions included hospital-based doula support, family members trained to serve as a lay doulas, and home visits; no included studies noted cultural concordance
	Postpartum: very satisfied with birth: 42/44 (95.1%) vs 85/97 (87.4%), aOR 2.43 (05% CI, 0.47–12.54)[47,63] 10 wk postpartum: overall very much agree that I feel satisfied with my birth: 6/8 ([calculated] 75%) vs 36/52 ([calculated] 69%), calculated RR 1.08, 95% CI, 0.70–1.68[47] 4 mo postpartum: gratification with labor and delivery: mean (SD) 16.9 (6.8) vs 16.9 (5.9), *P*=.993[60,61]	Very low	No benefit	
Low certainty of benefit or no benefit				
Adverse pregnancy outcomes				
HDP	Gestational hypertension: 22/187 (11.8%) vs 20/180 (11.1%), OR 1.07, 95% CI, 0.56–2.03	Very low	No benefit	RCT was conducted in women with low SES and reported cultural concordance of the doula and mother
	Preeclampsia after 20 wk: 42/486 (8.6%) vs 1,494/17,324 (8.6%), PS-adjusted RR 0.94, 95% CI, 0.70–1.26[64] 4/110 (3.6%) vs 73/1,463 (5.0%), calculated RR 0.73, 95% CI, 0.27–1.96[65] Gestational hypertension after 20 wk: 82/486 (16.9%) vs 3,478/17,324 (20.1%), PS-adjusted RR 0.87, 95% CI, 0.71–1.06[64] Hypertension (undefined): 14/110 vs 201/1,463, calculated RR 0.93, 95% CI, 0.56–1.54[65]	Low	No benefit	Hospital-based doula from a single institution[64,65]

(*continued*)

Table 3. Certainty of Evidence for Graded Outcomes (*continued*)

COE and Outcome	Summary of Findings (Doula vs Standard Maternity Care)	Grade	Direction of Effect	Applicability Considerations
Mode of delivery				
SVD	359/642 (55.9%) vs 333/620 (53.7%), pooled RD 0.056, 95% CI, −0.006 to 0.117, I^2: 43% (Appendix 9 Fig. 11)*	Low	Benefit	1 study reported that the sample included women with Medicaid and public insurance, and another study noted cultural concordance[11]; studies measured either the effect of doulas providing support during birth only or full-spectrum doula support that included medical–legal partnership with advocacy
	30/32 (93.2%) vs 58/73 (79.4), aOR 4.68, 95% CI, 1.14–19.28	Very low	Benefit	
VBAC	27/722 (3.8%) vs 12/722 (calculated 1.7%), PSM RR 2.48, 95% CI, 1.29–4.76[67] 22/64 (34.4%) vs 577/3,081 (18.7%), PS-adjusted RR, 1.83, 95% CI, 1.29–2.60[64]	Low	Benefit	1 study was hospital-based from a single institution with hospital-employed doulas[64]; the other was based on Medicaid claims from 9 states[67]
Use of obstetric intervention				
Epidural anesthesia	943/1,770 (53.3%) vs 1,146/1,749 (65.5%), pooled RD: −0.11, 95% CI, −0.16 to −0.07, I^2: 65% (Appendix 9 Figs. 12–13 for main and sensitivity analyses)*	Low	Benefit	Studies included women with low SES; 2 studies noted cultural concordance[11,12]; wide spectrum of doula services, ranging from labor-only support to prenatal, labor, and postpartum support plus medical–legal partnership with advocacy; rates in NRSIs were much lower than the national rate of 78%[74]
	703/2,279 (30.8%) vs 3,260/9,477 (34.4%), pooled RR 0.97, 95% CI, 0.87–1.07, I^2: 0% (Appendix 9 Fig. 14)*,♯	Very low	No benefit	
Labor outcomes				
Overall length of labor; length of 2nd stage of labor	3 of 4 trials reported a statistically significant difference: 7.4 h vs 8.4 h, calculated *P*=.01, N=412[40]; 10.4 h vs 11.7 h, *P*=.004, N=600[42]; the 3rd study[45] reported substantially shorter durations of latent, active, and transition labor (N=25) 1 trial reported no statistically significant difference but did not report sufficient information to permit pooling (N=221)[43] 3 trials reported no statistically significant differences for length of 2nd-stage labor but did not present sufficient information to permit pooling[38,42,43,45]	Low for overall length of labor and length of 2nd stage of labor	Benefit for overall length of labor; no benefit for 2nd stage of labor	2 of 4 RCTs noted a majority with low income[40,43]; cultural concordance was not noted in any study, but 1 had female friend or family member serve as lay doula after two 2-hour training sessions[42];the other studies assigned doulas to participants in labor

(*continued*)

Table 3. Certainty of Evidence for Graded Outcomes (*continued*)

COE and Outcome	Summary of Findings (Doula vs Standard Maternity Care)	Grade	Direction of Effect	Applicability Considerations
Perinatal outcomes				
Prenatal mood	Helped me manage my depression and anxiety: 260/1,396 (18.6%) vs 763/6,876 (11.1%), aOR 1.35, 95% CI, 1.13–1.62[66]	Low	Benefit	Mostly nulliparous participants who were in a private health plan and had access to a digital platform
Postpartum outcomes				
Initiation of breastfeeding or pumping	515/595 (86.6%) vs 479/592 (80.9%), pooled RD 0.04, 95% CI, −0.006 to 0.077, I^2: 44%; (Appendix 9 Figs. 15 and 16 for main and sensitivity analyses)*	Low	Benefit	1 RCT and the NRSIs were conducted in women with low SES[11,12,35,62,63]; 2 of 4 studies reported cultural concordance of the doula and the mother[11,35]
	3 high-ROB studies (N=11,837): pooled OR 1.22, 95% CI, 0.94–1.60, I^2: 27%, k=3,[12,57,62,63] (Appendix 9 Fig. 17)*,¶ Single moderate-ROB study demonstrated statistically significant benefit: OR 1.1, 95% CI, 1.08–1.16[57]	Low	Benefit	
Health care access and use				
Childbirth education attendance	64/128 (50.0%) vs 12/126 (9.5%), aOR 9.82, 95% CI, 4.84–19.89	Low	Benefit	Study recruited young mothers in communities with high levels of poverty and offered home visits during the last half of pregnancy and for 6 wk postpartum
Maternal readmission within 30 d postpartum	4/143 (2.8%) vs 3/143 (2.1%), aOR 1.53, 95% CI, 0.33–7.21	Very low	No benefit	Study recruited young mothers in communities with high levels of poverty and offered home visits during the last half of pregnancy and for 6 wk postpartum
Maternal readmission within 6–12 wk postpartum	67/1,318 (5.1%) vs 920/19,509 (4.7%), pooled RR 1.08, 95% CI, 0.81–1.43, k=3, I^2: 0%[64,65,67] (Appendix 9 Fig. 18)*	Low	No benefit	2 studies were hospital-based from a single institution with hospital-employed doulas[64,65]; the other was based on Medicaid claims from 9 states[67]
Patient-reported outcomes				
Reports of respectful care (including feeling heard, autonomy, informed refusal)	162/326 (49.6%) vs 715/936 (43.3%), partially adjusted OR 1.4, 95% CI, 1.0–1.8	Low	Benefit	Half of participants had public insurance; type, duration, and cultural concordance of doula support was not specified

(*continued*)

Table 3. Certainty of Evidence for Graded Outcomes (*continued*)

COE and Outcome	Summary of Findings (Doula vs Standard Maternity Care)	Grade	Direction of Effect	Applicability Considerations
Satisfaction with care (doula or health care team, received excellent care or high levels of support)	87/101 (86.0%) vs 68/120 (56.7%), RR 1.52, 95% CI, 1.28–1.81	Low	Benefit	1 of 2 studies had majority with low income[43]; the intervention was virtual doula support[50]; neither study noted cultural concordance, and both appeared to assign doulas to women
	733/1,396 (53.0%) vs 2,133/6,876 (31.0%), aOR 1.99, 95% CI, 1.74–2.28	Low	Benefit	
Self-efficacy (including confidence in navigating health care system)	MEQ 10 wk postpartum: mean (SD) 35.75 (3.94) vs 34.63 (3.36), calculated mean difference −1.12, 95% CI, −3.73–1.49[47] Survey conducted after birth: helped to decide birthing preference: 447/1,396 (32%) vs 907/6,876 (13%), aOR 2.45, 95% CI, 2.11–2.84[50,66] Helped to learn medically accurate information: 903/1,396 (67%) vs 3,285/6,876 (48%), aOR 1.47, 95% CI, 1.28–1.69[50,66] Confidence in ability to cope with tasks of motherhood: mean (SD) 20.2 (4.2) vs 22.8 (5.5), P=.152[60,61]	Low	Benefit	Populations included those with public and private insurance; 1 study was a virtual platform[50]; none mentioned cultural concordance
Parenting confidence	Average of 48 d postpartum Mean score (lower is better): Ability to be a good mother: 1.4 vs 1.8; t (df) 6.3 (492), P<.001[38] Senses baby's needs very well: 107/229 (47%) vs 74/265(28%), calculated RR 1.67, 95% CI, 1.32–2.12[38] Feels very well about managing the baby: 100/229 (44%) vs 76/265(29%), calculated RR 1.52, 95% CI, 1.20–1.94[38] 4–6 wk postpartum Labor had a very positive effect on feelings about ability to be a good mother: 71/143 (49.6) vs 67/145 (46.5), NS,[41] calculated RR 1.07, 95% CI, 0.84–1.37	Low	Benefit	1 study assigned doulas in labor for birth support only[41]; 1 trained friends or family members for 2 h to serve as lay doulas[38]; neither study noted cultural concordance or a focus on women with low income
Postpartum adjustment	Average of 48 d postpartum Very easy: 57/229 (25%) vs 39/265 (15%) 39/265 (15%), calculated RR 1.69, 95% CI, 1.18–2.44	Low	Benefit	Intervention was conducted in a population with majority low income; cultural concordance was not noted; doulas were assigned to participants once they were in labor

(*continued*)

Table 3. Certainty of Evidence for Graded Outcomes (*continued*)

COE and Outcome	Summary of Findings (Doula vs Standard Maternity Care)	Grade	Direction of Effect	Applicability Considerations
Infant outcomes				
NICU admission	21/142 (14.8%) vs 23/144 (16.0%), OR 0.87, 95% CI, 0.45–1.68[35] 24/187 (12.8%) vs 22/180 (12.2%), OR 1.06, 95% CI, 0.57–1.96[11]	Very low	No benefit	RCTs included home visiting by community-based doulas plus medical–legal support for nulliparous, lower-risk pregnant women with public insurance and a family support worker for women in high-poverty communities
	77/2,808 (2.7%) vs 574/17,226 (3.3%), pooled OR 0.82, 95% CI, 0.64–1.05, I^2: 0 RR from single moderate ROB study consistent with pooled results, PS-adjusted OR 0.84, 95% CI, 0.67–1.05[64] (Appendix 9 Fig. 19)*	Low	No benefit	1 NRSI included doula support for incarcerated women in 1 Midwestern state prison[48]; 2 were hospital-based doulas[45,64]
Infant mortality	0/156 (0%) vs 1/156 (1.29%), calculated Peto OR 0.135, 95% CI, 0.003–6.82	Very low	Benefit	RCT included home visiting by community-based doulas and a family support worker for young women in high-poverty Illinois communities.
	19/3,358 (5.79/1,000) vs 489/72,484 (6.75/1,000), predicted average treatment effect −2.66 percentage points, 95% CI, −3.18 to −2.14	Low	Benefit	NRSI included all live births from a single institution (University of Pittsburgh Medical Center)

COE, certainty of evidence; ROB, risk of bias; RCT, randomized controlled trial; OR, odds ratio; SES, socioeconomic status; NRSI, nonrandomized study of interventions; k, number of studies; RD, risk difference; PSM, propensity score matching; RR, risk ratio; PS, propensity score; aOR, adjusted odds ratio; HDP, hypertensive disorders of pregnancy; SVD, spontaneous vaginal delivery; VBAC, vaginal birth after cesarean; MEQ, Maternal Self-Efficacy Questionnaire; NS, not significant; NICU, neonatal intensive care unit.
* Appendix 9 is available online at https://links.lww.com/AOG/E807.
† In comparison, the pooled OR from RCTs was 0.75 (95% CI, 0.56–1.01).
‡ Most studies did not report rates of SVD or instrumented vaginal delivery. For RCTs, when SVD frequency was not reported but instrumental vaginal and cesarean delivery were, we inferred SVD results by subtracting frequencies of instrumental and cesarean deliveries from the overall number of deliveries.[11,44] Results from two high-ROB RCTs[40,41] that specifically reported SVD outcomes alone were imprecise, with wide CIs crossing the null.
§ For the single moderate ROB NRSI that reported only instrumental vaginal and cesarean deliveries, [57] we did not infer data for SVDs because we would not have been able to calculate adjusted effect estimates to address confounding without access to raw data.
‖ One NRSI (N=52,790)[24] was excluded from the pooled analyses because the breastfeeding data had the potential for observer or reporting bias (doulas reported initiation in the intervention arm), resulting in very high rates of breastfeeding initiation (97.9% vs 80.8%, calculated OR 11.31, 95% CI, 7.41–17.26).
¶ In comparison, the pooled OR from RCTs was 1.70 (95% CI, 1.19–2.43).
# In comparison, the pooled RR from RCTs was 0.85 (95% CI, 0.76–0.95).

Two moderate-ROB NRSIs reported statistically significant adjusted relative risk of attending the postpartum visit, ranging from 1.09[64] to 1.46[67] at 6–12 weeks postpartum. The size of the effect estimates was not consistent, but the results were precise, with no serious study limitations, leading to moderate certainty of benefit. No RCT reported this outcome.

We graded 16 outcomes from RCTs or NRSIs as having low certainty of benefit or no benefit of doula support compared with control interventions. Appendix 10 includes detailed results (https://links.lww.com/AOG/E807). Briefly, doula support was associated with more frequent spontaneous vaginal birth (low certainty of benefit from four RCTs[11,40,41,44] and very low certainty of benefit from one NRSI[47,62,63]), more frequent initiation of breastfeeding (low certainty of benefit from four RCTs[11,35,37,41] and low certainty of benefit from three NRSIs[12,57,62,63]), reduced infant mortality (very low certainty of benefit from one RCT[35] and low certainty of benefit from one NRSI[68]), and greater satisfaction with care (low certainty of benefit from one RCT[43] and one NRSI[50]). Doula support was associated with decreased epidural anesthesia (low certainty of benefit) in nine RCTs[11,35,38–44,46]; however, three NRSIs[12,47,57] yielded very low certainty of no benefit.

Doula support was not associated with reductions of hypertensive disorders of pregnancy (very low certainty of no benefit from one RCT[11] and low certainty of no benefit from two NRSIs[48,64]), maternal readmissions to the hospital (very low certainty of no benefit from one RCT[35] and low certainty of no benefit from three NRSIs[64,65,67]), and neonatal intensive care unit admissions (very low certainty of no benefit from two RCTs,[11,35] and low certainty of no benefit from three NRSIs[45,48,64]). For all other outcomes, studies reported benefits, but we found low certainty from either RCTs or NRSIs (vaginal birth after cesarean [k=2],[64,67] overall length of labor [k=4],[38,40,42,45] prenatal mood [k=1],[66] childbirth education attendance [k=1],[35] respectful care [k=1],[51] self-efficacy [k=3],[47,60,61,66] parenting confidence [k=2],[38,41] and postpartum adjustment [k=1][38]) (Table 3). Across all outcomes, reasons for low certainty include imprecision (few events, effect estimates with wide CIs or CIs spanning the null); high ROB; and, in some instances, inconsistency in effects.

Fifteen outcomes–instrumental vaginal delivery, Apgar scores, maternal self-esteem, postpartum mood disorders, fetal death, postpartum hemorrhage, use of oxytocin [before labor in one study,[45] in first stage of labor in one study,[41] timing unspecified in four studies], hospital length of stay, infant hospital readmission, severe maternal morbidity, episiotomy, labor induction, lactation, small-for-gestational-age birth weight or fetal growth restriction, and unexpected complications in term newborns–were graded as having very low certainty of benefit or no benefit for doula support compared with control interventions (Appendix 10, https://links.lww.com/AOG/E807); 10 of these outcomes were evaluated by two or fewer studies. Across all these outcomes, lack of precision, lack of consistency, and high ROB served to limit confidence in the results, resulting in very low certainty. Appendix 9 (available online at https://links.lww.com/AOG/E807) provides further details on results from individual studies and pooled estimate of effect when possible (for instrumental vaginal delivery and oxytocin use) for outcomes rated as having very low certainty of evidence. Appendix 10 (available online at https://links.lww.com/AOG/E807) lists 44 outcomes for which we found no gradable evidence.

Several meta-analyses (preterm birth, cesarean, spontaneous vaginal delivery, instrumental vaginal delivery, epidural anesthesia, and use of oxytocin) showed moderate or high I^2 values, suggesting that variability in the observed effects reflects true differences rather than sampling error. Although the number of studies was not large enough to assess the causes of heterogeneity formally, variations in baseline risks (control group event rates), comorbidities, study setting (single centers vs large national databases), insurance coverage (Medicaid only vs mixed insurance), and duration and intensity of doula support may partly explain the heterogeneity.

Three RCTs (two moderate ROB,[11,37] one high ROB[38]) and five NRSIs (one moderate ROB,[67] four high ROB[47,48,59,65]) were conducted solely in populations at risk of disparities but did not necessarily report on the same outcomes, limiting our observations to general trends. Two studies were conducted among women who identified as African American[37] or Black.[65] One moderate-ROB RCT reported a higher rate of initiation of breastfeeding among women receiving doula support[37]; one high-ROB NRSI reported statistically significant differences in the rate of postpartum visits, but not in prenatal visits, hypertensive disorders of pregnancy, maternal readmission, or cesarean delivery.[65]

One high-ROB NRSI conducted in carceral settings found no differences between women receiving doula support and the comparison group on rates of cesarean delivery, preterm birth, neonatal intensive care unit admission, and Apgar scores.[48]

Five studies (one moderate-ROB RCT,[11] one high-ROB RCT,[38] one moderate-ROB NRSI,[67] two high-ROB NRSIs[47,59]) were conducted solely in populations with Medicaid insurance. Of these, four reported at least some benefits for the group receiving doula support compared with the comparison group in measures of cesarean delivery, postpartum depression or anxiety[67]; birth experience, self-esteem, parenting confidence, and postpartum adjustment[38]; spontaneous delivery and birth experience[47]; and breastfeeding initiation.[59]

Of the 25 studies with mixed populations, eight reported stratified results for populations that experience disparities (detailed results in Appendix 11, available online at https://links.lww.com/AOG/E807). These studies generally did not report consistent differences in outcomes by race, age, or public insurance status.[11,12,49–51,59,64,66–68] None explicitly indicated that these subgroup analyses were adequately powered or prespecified. Five studies provided context for their race-based analyses,[11,49–

[51,59,66,67] and three did not. Studies varied in how they reported race (race and ethnicity combined, race and ethnicity reported separately, race and ancestry reported separately) but generally did not provide commentary on their definitions. In one exception, authors described differences in racial-ethnic categorization by data source.[59] Studies did not consistently report similar direction of effects among Black participants compared with the entire sample. Chance findings may have occurred due to very small event rates and sample sizes in these analyses.

No studies compared outcomes by timing of doula support. One NRSI with moderate ROB reported on the number of doula interactions, comparing the effects of one compared with two virtual doula appointment.[50,66] The study did not report formal dose-response analyses but did report progressively larger effect estimates for several outcomes (having a vaginal birth, deciding birth preference, receiving high levels of support, learning medically accurate information, and managing anxiety or depression) for two or more appointments compared with one appointment; CIs for these effect estimates overlapped. An analysis limited to Black participants in this population found statistically significantly lower rates of cesarean birth and improved depression or anxiety for two or more doula appointments but not for one appointment, suggesting potential benefit at higher intensity.

We found no evidence for any other aspects of intensity of care and for other doula care features. Specifically, we sought but did not find data comparing doula training, skills, and experience; doula type; doula employer; mode of interaction; and race, language, or cultural concordance.

DISCUSSION

In this systematic review and meta-analysis, we found that most of the available evidence addresses the outcomes of doula support compared with standard care. We found no evidence regarding the timing of doula support and very limited evidence on doula characteristics. The evidence base is marked by heterogeneity and absent or weak evidence for many outcomes. Similarly, the majority of published studies have high ROB and led to moderate or low certainty of benefit and low certainty of no benefit for some outcomes.

Doula support is likely to be associated with lower rates of preterm and cesarean birth, improved birth experience, and higher postpartum office visit attendance rates (moderate certainty of evidence). Doula support may be associated with benefit for prenatal mood, less use of epidural anesthesia, decreased length of labor, spontaneous vaginal birth, vaginal birth after cesarean, initiation of breastfeeding, childbirth education attendance, patient perception of receipt of respectful care, satisfaction with care, self-efficacy, parenting confidence, and postpartum adjustment to parenthood. Doula care does not appear to be associated with reduced hypertensive disorders of pregnancy, maternal readmissions, and neonatal intensive care admissions (low certainty of evidence). Our findings are consistent with those of a recent meta-analysis that found a decrease in cesarean birth with doula support during labor.[69]

Doula support is commonly framed as an intervention with the potential to address disparities in health outcomes, particularly maternal morbidity and mortality.[10,13,70,71] Our review found some information specific to disparities. Specifically, only one study reported on overall severe maternal morbidity and mortality, did not find a benefit, and did not report outcomes stratified by race or ethnicity.[67] Thirteen studies in the review were conducted in or reported stratified analyses for populations at risk of disparities. Variations in study design, ROB, and outcomes precluded clear conclusions about whether or how these interventions can specifically reduce disparities in health outcomes. Our findings suggest that the effects of doula support on birth experience, satisfaction with care, and respectful care could form the basis of benefits reported for clinical outcomes, but the extent to which these outcomes can help reduce disparities in maternal outcomes is yet to be fully elucidated. Many of the studies in this review recruited a diverse study population but did not perform adequately powered or preplanned stratified analyses to better address the role of doula support specific to populations at risk of disparities. Mixed methods studies could deepen our understanding of how doula support affects health outcomes. However, the predisposing factors for disparities, including structural biases,[72] often occur years and even generations before the adverse outcome; therefore, it is unrealistic to expect that simply providing doula support during pregnancy would be able to mitigate all of these factors.

The effect of doula support on cesarean births may be different in populations with higher or lower baseline rates of cesarean births than the populations included in this review. Mean baseline cesarean rates of 19% in the included trials are lower than national rates, which are more than 32% overall according to vital records data.[73] Adding doula support to these study settings may have had a smaller influence on cesarean rates than in settings with higher baseline cesarean rates where opportunities for prevention may be greater.

A key strength is that our multidisciplinary team included practicing doulas and an obstetrician, thus providing clinic-based and practice-based context when interpreting study results. The exclusion of doula support interventions outside the United States limits our ability to comment on other models of doula support. Similarly, the exclusion of cost outcomes limits our ability to provide cost-effectiveness evidence relevant to implementation or scale-up of doula support. Generalizability may be limited given the baseline heterogeneity of outcomes such as cesarean delivery and preterm birth in different populations.

The study's limitations are primarily related to the available evidence. Many outcomes of interest to stakeholders, particularly those related to partners, maternal health care access and use, and infant outcomes, were completely absent from the literature. Studies of doula care interventions were commonly set in hospitals and did not report the inclusion of private practice doulas. Additionally, homebirths were entirely absent from the evidence. As a result, current literature may not be entirely representative of current doula care in the United States. Information about the relative effectiveness of doula programs of different intensity was also lacking. A majority of included studies had high ROB, limiting certainty in their findings. Variations in definitions of doula support and outcome measures contributed to heterogeneity.

In conclusion, available research suggests that doula support is associated with improvement in some maternal outcomes, although many studies were rated as having high ROB. As momentum for expanding doula services gathers, research on type, intensity, and timing of doula support may better elucidate the potential of doula care to improve maternal and child outcomes and reduce health disparities.

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